Roll No.: _____

Amrita Vishwa Vidyapeetham

Int MSc/BSc Degree (Supplementary) Examination – July 2025

Second Semester

Integrated MSc Physics with Minor in Scientific Computing/Data Science

22PHY114, -Electricity and Magnetism

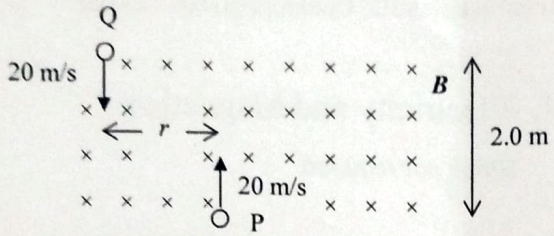
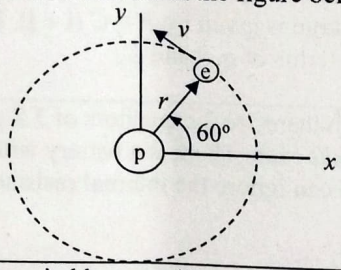
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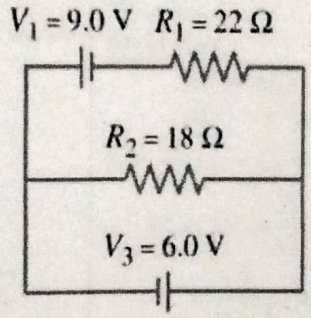
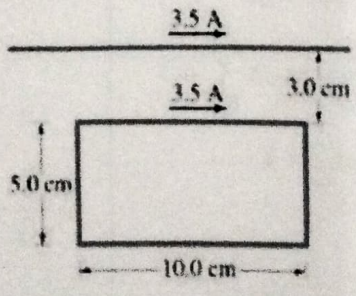
Time :3 hours Maximum Marks:100

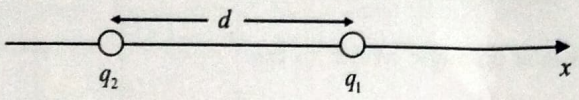
Instruction

1. Answer all the questions.
2. Answer with appropriate significant figures and units.

Qn No.		Marks	CO	Blooms Taxonomy level
1	A positive charge q travels in the direction of $+z$ with a speed of v. It then enters a region that has a uniform magnetic field $\boldsymbol{B}$. The magnitude of $\boldsymbol{B}$ is $3C/qv$ where C is a known positive constant. The magnetic force on the charge is given by $\boldsymbol{F} = C(\mathbf{i} + \mathbf{j})$. Find the three components of $\boldsymbol{B}$ in terms of q, v and C.	10	CO4	L3
2	There is series circuit in which there are 2 capacitors of 3.8 μF each and two resistors of 2.2 kΩ each. There is a battery with E = 12.0 V in the circuit. You can ignore the internal resistance of the battery. a) Determine the equivalent resistance in the circuit. b) Determine the equivalent capacitance in the circuit. c) How long does it take the current to drop from its initial value to 1.5 mA? Given that $I = (E/R)e^{(-t/RC)}$	10	CO3	L2, L3

3	P and Q are particles of mass 10^{-2} kg on the x-y plane. P has +0.20 C of charge and Q is electrically neutral. $\boldsymbol{B}$ is a 1.0 T uniform magnetic field directed perpendicularly into the page as shown in the figure below. At the same time, P and Q are projected into the magnetic field at 20 m/s.  a) What is the radius r of the circular path traversed by P? b) How long does P take to travel one quarter of a circle? c) What is the distance travelled by Q at the same time?	10	CO4	L3
4	The electron and proton in a hydrogen atom are separated by an average distance of 5.3×10^{-11} m. Assume the electron orbits with uniform circular motion as shown in the figure. a) What is the magnitude of the electric force on the electron? b) What is the electron's speed? c) If the electron's obit is on the x-y plane, express the electron's instantaneous velocity at the position shown in the figure below in terms of unit vectors.  	10	CO1	L3, L4
5	Each corner of a 1 m square is occupied by a positive charge, each of which has a magnitude q. Expressing your answer in terms of q and Coulomb's constant k, a) what is the electric field and electric potential at the centre of the square? b) what is the electric potential energy of any charge q? c) how much work is needed to remove one of these charges to infinity?	10	CO2	L2, L3

6	A square wire loop of sides 4.0 cm is in a uniform magnetic field B of magnitude 0.15 T. a) What is the magnetic flux when **B** is perpendicular to the face of the loop? b) What is the magnetic flux when **B** is at an angle of 60° to the area **A** of the loop? c) If the angle between **B** and **A** is changed from 60° to 0° in 0.10 s, what is the magnitude of the average current in the loop if its resistance is 0.12 Ω?	**10**	**CO5**	**L2, L3**
7	In an LR circuit, it takes 2.00×10^{-3} s to reach 50 % of the maximum value of the current. Given that $I = I_{max}(1 - e^{(-tR/L)})$ a) Determine the time constant of the circuit. b) Find the resistance if $L = 28.8 \times 10^{-3}$ H.	**10**	**CO5**	**L3**
8	In the figure, determine the currents I_1 and I_2 which flow through R_1 and R_2 respectively. 	**10**	**CO3**	**L3**
9	The figure shows a straight wire placed near a rectangular loop. The current in the wire as well as the loop is both 3.5 A. The different distances are also given in the figure. Find the magnitude and direction (i.e., towards the wire or away from the wire) of the net force on the loop. 	**10**	**CO4**	**L4**

10	Two-point charges $q_1 = 3.4 \times 10^{-6}$ C and $q_2 = -2.0 \times 10^{-6}$ C separated by a distance $d = 0.050$ m along the x-axis as shown in the diagram.  At what points along the x-axis, is the total electric potential zero?	**10**	**CO2**	**L3**

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create

**********End of Questions**********